

generalized_hill_cipher (w/ nice playfair, w/ nice binary)

S1 derive structure of non-square matrix from berlin clock

1 = u0	0 = circle	u = yellow
2 = nW nW nW nW	W = square	n = red
3 = nW nW nW nW	V = rectangle	
4 = uV uV nV uV uV nV uV uV nV uV uV		
5 = uW uW uW uW		

step 1 {substitute u0 -> nV}

```
1 nV
2 nW nW nW nW
3 nW nW nW nW
4 uV uV nV uV uV nV uV uV nV uV uV
5 uW uW uW uW
```

step 3 {transpose 3 <-> 4}&
{wrap 4 to width 3}

1 nW nW nW nW	4 uV uV nV
2 nW nW nW nW	5 uV uV nV
3 uW uW uW uW	6 uV uV nV
	7 uV uV nV

step 2 {concatenate 4 -> 4 1}

```
1 nW nW nW nW
2 nW nW nW nW
3 uV uV nV uV uV nV uV uV nV uV nV
4 uW uW uW uW
```

step 4 {partition 123 into matrix B}&
{partition 4567 into matrix E}

B	nW nW nW nW	E	uV uV nV
	nW nW nW nW		uV uV nV
	uW uW uW uW		uV uV nV
			uV uV nV

B=Substitute(E^T,{u->n, V->W})

^T is MatrixTranspose

E=Substitute(B^T,{n->u, W->V})

S2 derive the structure of the functions

nW*x = (noW)(x) = W(n(x))	n = u^-1	n,u,W,V are functions
uW*x = (uoW)(x) = W(u(x))	u = n^-1	^-1 is function inverse
nV*x = (noV)(x) = V(n(x))	W = V^-1	o is function composition
uV*x = (uoV)(x) = V(u(x))	V = W^-1	x is any string

S3 derive the structure of the input and output of the functions

plaintext	P3 = P31 P32 P33	P4 = P41 P42 P43 P44
ciphertext	C3 = C31 C32 C33	C4 = C41 C42 C43 c44

3 strings of 32 characters each 4 strings of 24 characters each

for example, we can define these c3 and c4 explicitly by partitioning k4 into 12 equal pieces of 8 characters each in four ways by removing singleton characters M or Y or both. For "Y" there are 64 characters to the left and 32 characters to the right. For "M" the divisions are different. see *page2 there are two characters whose count is 1 in k4: Y and M, if Y or M is removed, the ct has 96 characters and 25 different characters like 1-1 perfect playfair if both y and m are removed there are 24 different characters in k4, so it can be converted into 5bit perfect binary 1-1 mirror 1-1 complement see *page2

S4 define the structure of the relation from the ct to pt

the relation is only one of the four maps:

1 B.C4^T=>P3^T	3 B.P4^T=>C3^T	. is Matrix*	A must havegeneralized ^-1
2 E.C3^T=>P4^T	4 E.P3^T=>C4^T	A is (B or E)	A^-1 (Ax=b=>AA^-1b=b)

*ex

```

-----
                grid y
OBKRUOXO GHULBSOL IFBBWFLR VQQPRNGK
SSOTWTQS JQSSEKZZ WATJKLUD IAWINFBN [Y]
PVTMTZFP KWGDKZXT JCDIGKUH UAUEKCAR

c31 = OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGK
c32 = SSOTWTQSJQSSEKZZWATJKLUDIAWINFBN
c33 = PVTMTZFPKWGDKZXTJCDIGKUHUAUEKCAR

c41 = OBKRUOXOSSOTWTQSPVTMTZFP
c42 = GHULBSOLJQSSEKZZKWGDKZXT
c43 = IFBBWFLRWATJKLUDJCDIGKUH
c44 = VQQPRNGKIAWINFBNUAUEKCAR

```

```

                grid m
O B K      W A T
R U O      J K L
X O G      U D I
H U L      A W I
B S O      N F B
L I F      N Y P
B B W      V T T [M]
F L R      Z F P

V Q Q      K W G
P R N      D K Z
G K S      X T J
S O T      C D I
W T Q      G K U
S J Q      H U A
S S E      U E K
K Z Z      C A R

c31 = ORXHBLBFVPGSWSSKWJUANNVZKDXCGHUC
c32 = BUOUSIBLQRKOTJSZAKDWFYTFWKTDKUEA
c33 = KOGLOFWRQNSTQQEZTLIIBPTPGZJIIUAKR

c41 = OBKRUOXOGHULBSOLIFBBWFLR
c42 = VQQPRNGKSSOTWTQSJQSSEKZZ
c43 = WATJKLUDIAWINFBNYPVTMTZFP
c44 = KWGDKZXTJCDIGKUHUAUEKCAR

```

there are many other possibilities within these grids, such as rotations, here is one:

```

c41 = ORXHBLBFBUOUSIBLKOGLOFWR
c42 = VPGSWSSKQRKOTJSZQNSTQQEZ
c43 = WJUANNVZAKDWFYFTFLIIBPTP
c44 = KDXCGHUCWKTDKUEAGZJIIUAKR

```

grid m y

```

OBKRUOXOGHULBSOLIFB
BWFLRVQQPRNGKSSOTWT
QSJQSSEKZZWATJKLUDI
AWINFBNPVTMTZFPKWGDK
ZXTJCDIGKUHUAUEKCAR

```

1-1 playfair

no y	no m
K R P T O	K R Y P T
S A B C D	O S A B C
E F G H I	D E F G H
J L M N Q	I J L N Q
U V W X Z	U V W X Z

24->2^5 perfect binary 1-1 distinct mirror 1-1 distinct complement

bit	# bit	AB	t	AB	2	0	1	shift from center
					+	-	+	direction
1	01	K	00011	-11000	A	01110	G	00110--01100 Q
32	10	R	11100	--00111	B	10001	H	11001--10011 U
2	01	P	00101	--10100	C	01010	I	01001--10010 V
	10	T	11010	--01011	D	10101	J	10110--01101 W
41	01	O	00001	--10000	E	00100	L	00010--01000 X
1	10	S	11110	--01111	F	11011	N	11101--10111 Z
50	01					00000		
1	10					11111		

* center not used, for shifts 2+ is inverse with 1+ and 2- is inverse with 1-